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Informal Work in India: A Tale of Two Definitions

Rayees Ahmad Sheikh¹ · Sarthak Gaurav¹

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Abstract

Despite growing informality in developing economies, identifying correlates of informality continues to be a challenge due to multiple definitions of informality as well as data limitations. In order to explain the determinants of informality, the authors use two operational definitions of informality, namely ‘informal sector’ and ‘informal employment’, based on enterprise characteristics and employment characteristics, respectively. Using unit-level data from a nationally representative dataset, the authors find that, irrespective of how informality is defined, workers’ education, vocational training and gender play a significant role in determining participation in the informal labour market. The results of this study are robust to correction for selection bias and controlling for regional variations. The findings emphasise the need to restructure skill development programmes to account for the heterogeneity of informal workers.

Keywords Informal sector · Informal employment · Selection bias

Résumé

Malgré l’augmentation du travail informel dans les économies en développement, l’identification des corrélats de l’informalité continue d’être un défi en raison des définitions multiples de l’informalité ainsi que des limites des données. Afin d’expliquer les déterminants du travail informel, les auteurs utilisent deux définitions opérationnelles de l’informalité, à savoir «secteur informel» et «emploi informel» selon les caractéristiques respectives de l’entreprise et de l’emploi. Grâce aux données unitaires provenant d’un ensemble de données représentatives au niveau national, les auteurs constatent que, quelle que soit la définition de l’informalité, l’éducation des travailleurs, la formation professionnelle et le sexe jouent un rôle important dans la détermination de la participation au marché du travail informel. Les résultats de cette étude sont robustes quant à la correction du biais de sélection et au contrôle des vari-

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ations régionales. Les résultats soulignent la nécessité de restructurer les programmes de développement des compétences pour prendre en compte l'hétérogénéité des travailleurs informels.

A large share of the workforce in the developing world is outside the purview of legislation and excluded from social security benefits (Huitfeldt and Jütting 2009; Schneider 2012; La Porta and Shleifer 2014). Despite increasing attention in policy and academic discourse, measuring the incidence and causes of informality remains a challenge. This is primarily due to the prevalence of multiple definitions and ambiguity in the conceptualisation of informality (Kanbur 2009; Henley et al. 2009). To address these measurement and conceptualisation issues, two definitions of informality were adopted by the International Labour Organization (ILO), viz. 'informal sector' and 'informal employment', which covers two different aspects of informality that have gained prominence. While the former focusses on firm or enterprise characteristics, the latter takes into account the employment conditions of the worker. Furthermore, the definitions are often tempered by data availability in operationalising informality (Lehmann 2015).

In this paper, we study informality in the Indian labour market by using unit-level data from a nationally representative sample comprising approximately half a million individuals for the year 2011–2012.¹ Our intent is to examine whether determinants of informality differ by the definition followed, and assess the implications of such differences. After controlling for selection bias, we find that females, urban residents and more educated workers were less likely to be employed informally.² Furthermore, workers with formal skills training were less likely to have informal work. The study findings are robust to regional variations.

Evidence on the correlates of informality examined from an enterprise as well worker's perspective is crucial for developing economies, such as India, on the following grounds: First, fast-growing emerging economies that are experiencing a demographic transition are characterised by a large informal sector. For instance, the informal sector in India accounted for 82% of the workforce in 2011–2012 (ILO 2016). Second, the developmental consequences of informality are considerable, given the characteristics of those engaged in informal work. Workers with employment in the informal sector or informal employment are typically less educated, and most belong to socially and economically disadvantaged communities. More often than not, these jobs are characterised by exclusion from benefits of social security and fail to have decent work conditions. Without binding legal contracts, workers are vulnerable to spells of unemployment and poverty (Standing 1999; Huitfeldt and Jütting 2009; Breman 1996). Third, the phenomenon of 'jobless growth' (World Bank 2018; Thomas 2014; Mehrotra et al. 2015; Himanshu 2011; Kannan and

¹ Unit-level data pertains to data from the ultimate stage of the large scale multi-stage sample survey: household level. The non-agricultural sector comprises enterprises engaged in mining and quarrying, fisheries, and enterprises in the manufacturing and service sectors.

² Informality or informal work refers to both informal sector employment and informal employment.



Raveendran 2009) in the formal sector, particularly the formal manufacturing sector, makes understanding of informal employment critical. Furthermore, despite rapid economic growth, most of the jobs being created are informal in nature.

This study contributes to an improved understanding of determinants of participation in informal work in developing and emerging economies such as India. Since the informal sector is defined on the basis of enterprise characteristics while informal employment takes into account workers' characteristics, the trend of growing informality can be examined from two perspectives: either an increase in employment in informal enterprises or an increase in workers having informal employment. Indeed, workers in the informal sector are less likely to receive social security benefits, while those in the formal sector may also be bereft of social security benefits. In this context, this study brings out a lesser-known facet of labour market dynamics in the developing world, i.e., formal sector growth can coexist with growth of informal employment. Despite economic growth, growing informality is a major policy concern, and identifying strategies to target the informal sector or informal employment is crucial. Therefore, understanding the determinants of informality has potential policy implications, particularly in identifying vulnerable groups in the labour market and examining the effectiveness of skills and training initiatives.

This paper is organised as follows: First, the fundamental concepts and definitions of informality are discussed. The important aspects of informality and contextualisation of informality in India are explained. A brief review of the literature on informality and its determinants is presented, followed by a description of the data, analytical model and empirical strategy applied. The researchers then present and discuss the results of the study, followed by their conclusions.

Informal Sector Employment and Informal Employment: Concepts and Definitions

The term 'informal sector' was introduced into the economics literature by Hart (1973) in order to distinguish between formal and informal income opportunities, for instance wage-earning and self-employment. Since then, different criteria have been used to identify the informal sector, but none of them are universally applicable in different empirical situations (Papola 1981). The following definition distinctions are central to an improved understanding of informality.

Employment in the Informal Sector and Informal Employment

The informal sector and informal employment are two aspects of informality that complement one another in explaining the state of the labour market. While the former takes into account the informality of enterprises, the latter looks into the informality of jobs. The 15th International Conference of Labour Statisticians (ICLS) defined the informal sector as private unincorporated enterprises producing goods or services meant for sale or barter, having size, in terms of number of employees, below a certain threshold, and that are engaged in non-agricultural activities



(Husmanns 2001, 2002).³ According to this definition, informal sector employment comprises all jobs in informal sector enterprises. It is worth noting here that agricultural workers are excluded from the informal sector as per this definition (Husmanns 2004, p. 3).

While informal sector employment is defined at enterprise level, informal employment is defined at individual or worker level. As per the 17th ICLS, informal employment comprised all ‘informal jobs’—carried out in formal sector enterprises or informal sector enterprises or households during a given reference period (Husmanns 2004).⁴ This implies that informal jobs can be present in both the informal sector and the formal sector. Informal jobs are characterised by the absence of job contracts, social protection, entitlement to certain employment benefits, and not being subject to labour legislation or taxation.

Furthermore, the ILO definition of informal sector suggests that the threshold number of employees in the definition should be decided as per national circumstances. In India, the National Commission for Enterprises in the Unorganised Sector (NCEUS) specifies this threshold number of employees for the classification of enterprises as being in the informal sector. Alternatively, the Factories Act of 1948 classifies businesses with over 10 workers that use electricity for production, and businesses with 20 workers that do not use electricity for production as formal sector enterprises. However, the NCEUS adopted the term ‘informal sector’ for India and defined the informal sector as per ILO guidelines. Accordingly, we used the following definitions of informal sector and informal employment by the NCEUS:

1. Informal sector: ‘The informal sector consists of all unincorporated private enterprises owned by individuals or households engaged in the sale and production of goods and services operated on a proprietary or partnership basis and with less than 10 total workers’ (NCEUS 2008, p. 2).
2. Informal employment: ‘Informal workers consist of those working in the informal sector or households, excluding regular workers with social security benefits provided by the employers, and the workers in the formal sector without any employment or social security benefits provided by an employer’ (NCEUS 2008, p. 2).

In order to capture the dimensions of informality at enterprise level and worker level, we classify the informal sector into two mutually exclusive categories, namely informal sector informal employment (ISIE) and informal sector formal employment (ISFE). Accordingly, the formal sector is also classified into two mutually exclusive categories: formal sector informal employment (FSIE) and formal sector

³ The ICLS is an ILO meeting that takes place every 5 years. Participants include experts from governments, as well as from workers’ and employers’ organisations.

⁴ According to the 17th ICLS, ‘household’ includes households producing goods exclusively for their own final use (e.g. subsistence farming), as well as households employing paid domestic workers (maids, drivers etc.)



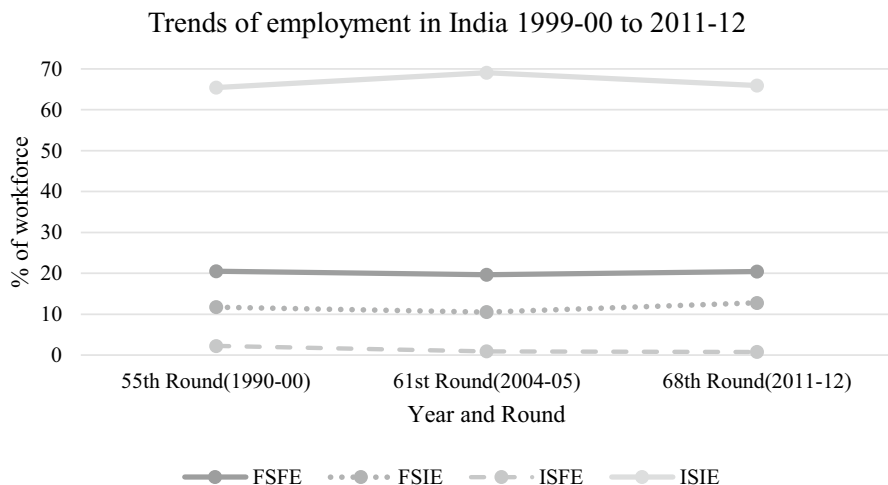


Fig. 1 Trends in employment in India from 1999–00 to 2011–12. FSFE, formal sector formal employment; FSIE, formal sector informal employment; ISFE, informal sector formal employment; ISIE, informal sector informal employment. *Source* Authors' own calculation based on 55th, 61st and 68th round of NSSO employment–unemployment survey, unit-level data

formal employment (FSFE). We summarise the trends in employment in India by examining these four categories of employment based on definitions of informality.

Figure 1 shows the trend of different employment types from 1999–2000 to 2011–2012, spanning three rounds of the National Sample Survey Office (NSSO) employment–unemployment survey, viz. the 55th, 61st and 68th rounds. It is evident that informality in India has been not only high but also persistent during this time period. Employment in ISIE was highest in all three NSS rounds at around 65%, whilst FSFE remained little below 20% in all three rounds. On the contrary, ISFE was minimal at around 1%, whilst FSIE remained above 10% and grew in the last two rounds. The growth in FSIE in the last two rounds indicates that large enterprises in the formal sector may be increasingly offering informal employment. This section of workers is equally vulnerable as ISIE sub-group in terms of provision of social security benefits, and security of employment. The trends reported in Fig. 1 do not include data from the 66th round of NSSO employment–unemployment survey 2009–10 as data from that round had concerns about validity. However, the 68th employment–unemployment round was the last round of employment–unemployment survey conducted by NSSO. It was replaced by the Periodic Labour Force Survey (PLFS) in 2017–18. PLFS data indicate similar trends of informality but we did not include the same here as two data sets are not strictly comparable.

Literature Review

A well-recognised fact about labour markets in developing countries is the presence of a large informal sector (Pradhan and Van Soest 1995; La Porta and Shleifer 2014). Various explanations have been put forth in literature to explain the persistence and growth of the informal sector (Maloney 1999, 2004; Fields 1990; De Soto 1989; Günther and Launov 2012; Magnac 1991). Some studies have explored informality from the perspective of the informal sector (Amaral and Quintin 2006; Azuma and Grossman 2008; Chaudhuri 1989), while others have investigated informality from an informal employment perspective (Günther and Launov 2012; Kucera and Roncolato 2008; Lehmann 2015; Sapkal and Sundar 2017).

Informal firms in developing countries are characterised as unincorporated, small size (in terms of number of employees) and predominantly household-run firms employing a large share of the workforce (La Porta and Shleifer 2014). Differences in wages and capital costs between the formal and informal sectors (Stark 1982; Chaudhuri 1989; Woodruff and McKenzie 2006), as well as credit constraints (Banerjee and Duflo 2008; Raj and Sen 2015), have been shown to impede the transition of small, informal firms to large formal firms. Furthermore, tax considerations and regulatory costs have also been propounded as a driver of a firm's choice to be in the informal sector (Basu et al. 2012; Demenet et al. 2016). Institutional factors, such as higher tax burden (De Soto 1989; Loayza 1996; Azuma and Grossman 2008) and bureaucratic hassles (Friedman et al. 2000), have been found to drive firms to the informal sector, especially in the case of poor countries. It has also been argued that cost cutting through subcontracting and outsourcing is one of the factors contributing to the growth of informality (Chen 2012; ILO 2014).

Examining determinants of informality has been a long-standing issue due to the prevalence of multiple definitions of informality (Lehmann 2015). As discussed above, two definitions of informality are primarily used, viz. the informal sector and informal employment, with the former focussing on enterprise characteristics and the latter on characteristics of employment. From the workers' perspective, in the context of the dichotomization of the labour market, there are two contested issues. The first issue is whether informal work is a voluntary choice of the worker (Kucera and Roncolato 2008), or if workers participate in informal work due to exclusion or 'push' factors (Maloney 1999; Fields 2003; Perry et al. 2007; Günther and Launov 2012). Second, it has also been argued that the choice of a worker to be in the informal sector or engage in informal employment may be motivated by a desire to avoid tax burdens and inefficient regulations (Fields 2003). Contrary to this, there is a view that informality is a means of capitalistic exploitation fostered by regulatory issues (Breman 1996). In this context, Maiti and Sen (2010) classify theories of the informal sector into two groups: 'means of exploitation' and 'means of capital accumulation'. However, their results show mixed evidence on whether there is exploitation or accumulation of capital in the informal sector.

Based on the two definitions of informality, some studies have attempted to explain the determinants of informality. Studies have found factors such as education, age, gender and place of residence (Bairagya 2012; Narayanan 2015; Sahoo



and Neog 2017) as significant in explaining participation in the informal sector. Low education has also been shown to be one of the most important factors in determining informal sector employment (Bairagya 2012; Abraham 2017). As jobs in the formal sector are scarce, informal sector work is mostly taken up by young workers with low educational attainment to avoid unemployment. In addition, it provides them with a stepping stone to the labour market (Maloney 2004). From a gender perspective, women have been found to be more likely to work in the informal sector compared with men, as women in developing economies generally have less education and skills (Chen 2001). Furthermore, cultural and social factors restrict women to household work, thereby limiting investments in female education and skills training (Chen 2001).

In terms of socio-economic determinants of informal work, Bairagya (2012) found that workers belonging to other backward classes (OBCs) are more likely to work in the informal sector. In another study, scheduled caste (SC) and scheduled tribe (ST) workers were found to be more likely to be in the informal employment sector, even in public enterprises (Kannan 2014). In the context of rural–urban migration among workers, an important aspect of informality has been the exclusion of ‘footloose labour’ (Bremen 1996) from social security benefits, as these workers move frequently between casual and insecure jobs in the informal sector. In a recent study, Sapkal and Sundar (2017) examined the determinants of informal employment from a contract of employment perspective: workers with no contract, workers with short-term contracts (less than a year) and workers with long-term contracts. They found that those with long-term employment contracts earned more and enjoyed social security benefits.

Now, given the theoretical views on informality, it is important to investigate the determinants of participation in informal sector employment and informal employment.

Data and Methodology

We used unit-level data from the 68th round of the employment–unemployment survey of NSSO. The survey was conducted in the year 2011–2012. It included approximately half a million individuals from 101,724 households across all states and regions of India, surveyed over the period from July 2011 to June 2012. The sample of interest for this analysis was the ‘workforce’, comprising individuals aged 15–60 years employed in non-agricultural activities.⁵ The data included details on

⁵ The reference periods used by the NSSO are: the usual principal activity status (UPAS) pertaining to the activity for the major part of a year with a reference period of the past 365 days; usual principal and subsidiary status (UPSS) relating to the status of the principal activity and subsidiary activities in the reference period of the previous 365 days; current weekly status (CWS), that is, the activity status pursued during a reference period of 7 days preceding the survey date; and the current daily status (CDS), that is, the time disposition of a person on each day of the reference week. Data in this study are from the last employment–unemployment survey of NSSO. Since 2017–18, the Periodic Labour Force Survey (PLFS) is the official source of information on employment and unemployment in India.



employment patterns, asset ownership, demography, job characteristics and the enterprise characteristics of workers. We did not include agricultural workers in the sample, in accordance with the ICLS definition.⁶ Furthermore, we dropped workers for whom information on enterprise type was missing, resulting in a reduction of the sample for analysis to 104,182 individuals.⁷

The measure of employment used in the study was usual principal activity status (UPAS).⁸ Although the data provided information on employment in subsidiary activities as well, we focussed on UPAS in order to understand informal work in primary activities. The working status of an individual (activity status) was also classified into four different job categories: (1) self-employed, (2) regular salaried workers, (3) casual wage-workers and (4) unpaid family workers. The self-employed category consisted of own-account workers and employers. Own-account workers were classified as informal, as they were not employed by any enterprise. Self-employed employers were classified under the informal sector as well, unless 10 or more workers were employed in the enterprise. However, self-employed workers could be classified as having formal or informal employment based on whether they were covered by social security benefits. As per this classification, more than half of workers were self-employed.

Analytical Model

In the analytical model, formal sector employment is the most desired employment choice. However, jobs are not allocated according to market forces. There are entry barriers to formal sector employment, but we posited employment in the informal sector as a voluntary choice. The reason for considering informal sector work as voluntary is that working in the informal sector suggests non-wage benefits, such as proximity to home and flexible work hours.

To model the employment choice of a worker, consider a labour market where workers differ in their finite set of observable characteristics, X . Workers can be employed in either the formal sector (FS) or the informal sector (IS). Both sectors offer finite jobs, but only some of the workers receive social security, the distribution of which, within a sector, is random. Therefore, workers within each sector will either have formal employment (FE) or informal employment (IE), depending on whether or not they receive social security benefits. Therefore, employment is viewed as four types: (1) FSFE—employed in the formal sector and receiving social security benefits, (2) FSIE—employed in the formal sector and not receiving social security benefits, (3) ISFE—employed in the informal sector and receiving social security benefits and (4) ISIE—employed in the informal sector and not receiving

⁶ We dropped individuals having national industrial classification (NIC) code 1, that is, those who are agricultural workers: including cultivators and agricultural labourers. However, we included workers in allied activities such as fisheries, mining and quarrying.

⁷ The observations dropped were 143,076 individuals based on UPAS and 41,524 agricultural workers.

⁸ Employment status based on a reference period of 365 days.



social security benefits. A worker's choice of employment can be posed as a two-stage process; that is, the new entrant chooses the sector in the first stage, and the employment type in the second stage.

A worker's probability of choosing the sector given her/his characteristics in the first stage is given as

$$p(S_l) = f(X_l, \theta), \quad (1)$$

where p is the probability of a worker l being employed in sector S (informal or formal), X denotes the worker characteristics, and θ represents the state of the economy.

In the second stage, the probability of being formally employed (with social security benefits) in the sector chosen in Eq. (1) is given as

$$p(E_l|S_l) = g(A(\theta)), \quad (2)$$

where E_l is formal employment for worker l in a sector, A is the availability of formal employment, i.e. the number of vacancies in any given sector with social security benefits, which is a function of the state of the economy, θ .

The intuition behind the choice of employment is better understood by assuming that there is a clear hierarchy in jobs along a wage continuum, with FSFE being associated with the highest wage and ISIE being associated with the lowest wage. Let W_i be the wage rate in the informal sector and W_f be the wage rate in the formal sector. Since formal employment provides social security benefits (s) in addition to wages, the effective wage rate in FSFE, FSIE, ISFE and ISIE is $W_f + s$, W_f , $W_i + s$ and W_i , respectively, such that $s < (W_f - W_i)$. The choice of employment by a worker is motivated by wage differential given the worker characteristics.

Estimation Strategy

In order to estimate the determinants of participation in the informal sector or informal employment, we used a multinomial logit model. However, in order to control for selection bias, we employed Heckman's two-stage model (Heckman 1976, 1979). In the first stage (selection model), we estimated a probit model of whether or not an individual participates in the labour market as represented by Eq. (1). The selection model is given by

$$P(I_i) = \alpha_i + \beta_{1i}X_{1i} + \beta_{2i}X_{2i} + u_{1i}, \quad (3)$$

where I_i is the dependent variable indicating whether or not individual i is employed in the informal or formal sector. X_{1i} is a vector of observable worker characteristics, such as gender, age, education and social group, that determine the probability of working in any sector. X_{2i} denotes the exclusion restriction, and u_{1i} is the error term.⁹

⁹ This is the ratio of dependants in a household to the total number of members of the household. The dependants in a household are children below 5 years of age and those above 60 years of age and not in the workforce.



Table 1 Description of variables

Variable	Description
Dependent variables	
Employment	Whether individual is in labour force = 1, otherwise = 0
Employment choice	Formal employment in formal sector = 1, informal employment in formal sector = 2, formal employment in informal sector = 3, informal employment in informal sector = 4
Explanatory variables	
Wages	Weekly wages and salaries in Indian rupees (INR)
Age	Age of individual (years)
Age squared	Square of individual's age
Education	Education of individual: base category not literate; 2 = literate without formal schooling, 3 = below primary, 4 = primary, 5 = middle, 6 = secondary, 7 = higher secondary, 8 = diploma/certificate, 9 = graduate, 10 = postgraduate and above
Land	Natural log of land owned by household (hectares)
Vocational training*	Dummy variable for vocational training = 1 if individual reported formal vocational training, 0 otherwise
Female	Dummy variable for female = 1 if individual is female, 0 otherwise
Dependency ratio (DR)	The ratio of dependants in a household to the total number of members of the household. Dependants in household are children below 5 years of age and those above 60 years of age, and not in workforce
Social group	Designates the social group an individual belongs to, namely Scheduled Tribe (ST), Scheduled Caste (SC), Other Backward Class (OBC), and others (base category)
Religion	Designates the religion an individual belongs to: Hinduism (base category), 2 = Islam, 3 = Christianity, 4 = Sikhism, 5 = Others
State	Control for 28 states and 7 union territories of India

*Vocational training includes only those who have received formal vocational training or were receiving formal vocational training at the time of survey

The exclusion restriction variable is included in the selection model, but excluded from the outcome model to ensure that selection into the labour market and staying out of the labour force do not follow from the same process, thereby enabling identification (Greene 2010). We use dependency ratio (as defined in Table 1) as the exclusion restriction, since the said variable does not determine the type of employment but impacts the selection decision.

From the selection Eq. (3), we obtained the correction factor λ (inverse Mills ratio) for use in the second stage (outcome model):

$$\Pr(I_i = j) = \gamma_i X_i + \lambda + u_{2i}, \quad (4)$$

where X_i is the vector of observable characteristics, such as gender, age and education, that determine participation, λ is the Inverse Mills Ratio (IMR), and u_{2i} is the error term. We estimated the outcome model using a multinomial logit regression.



Table 2 Descriptive statistics of continuous variables *Source* Authors' own calculations

Variable	Workforce			Formal sector			Informal sector			Formal employment			Informal employment		
	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N
Age	36.44	10.67	104,182	39.21	10.32	25,897	35.52	10.57	66,104	40.82	9.92	20,622	35.33	10.51	71,379
Wage earnings	2508	4067	54,432	3875	5516	25,218	1337	1190	24,111	4575	5938	20,347	1272	1065	28,977
Land (ha)	0.414	1.22	88,498	0.55	1.40	21,072	0.30	1.00	56,374	0.587	1.46	16,729	0.313	1.01	60,717

1 ha = 2.47 acre

N number of observations for each variable, SD standard deviation



Table 3 Descriptive statistics of categorical variables

	Workforce (%)	Formal sector (%)	Informal sector (%)	Formal employment (%)	Informal employment (%)
Vocational training	5.65	10.22	4.51	11.18	4.66
Education					
Not literate	15.35	5.52	16.73	5.52	16.73
Literate without formal schooling	0.36	0.19	0.37	0.19	0.37
Below primary	7.86	3.07	9.14	3.07	9.14
Primary	12.13	5.21	14.20	5.21	14.20
Middle	18.38	11.11	21.28	11.11	21.28
Secondary	15.15	14.68	16.18	14.68	16.18
Higher secondary	10.52	14.98	9.44	14.98	9.44
Diploma/certificate	2.56	5.55	1.70	5.55	1.70
Graduate	12.82	26.97	8.56	26.97	8.56
Postgraduate and above	4.87	12.73	2.39	12.73	2.39
Gender					
Male	80.96	79.72	84.29	79.72	84.29
Female	19.04	20.28	15.71	20.28	15.71
Sector					
Rural	50.64	44.68	48.54	55.32	51.46
Urban	49.36	55.32	51.46	44.68	48.64
Social group					
Others (unreserved)	32.47	37.46	32.25	39.00	32.19
Scheduled tribe (ST)	12.33	17.92	8.67	18.53	9.17
Scheduled caste (SC)	16.37	14.25	16.42	12.52	16.76
Other backward classes (OBC)	38.83	30.37	42.66	29.95	41.72



Table 3 (continued)

Religion	Workforce (%)	Formal sector (%)	Informal sector (%)	Formal employment (%)	Informal employment (%)
Hinduism	74.46	73.81	75.06	74.12	74.30
Islam	14.65	17.46	9.54	17.05	8.95
Christianity	6.99	5.05	10.91	5.10	12.23
Sikhism	1.86	2.07	1.72	2.07	1.63
Others	2.05	1.61	2.76	1.65	2.89
Observations	104,181	25,897	66,104	25,893	71,378

All estimates in the table are unweighted



The errors of Eq. (3) and (4) are correlated and jointly follow the bivariate normal distribution (Huesca and Camberos 2010).

Sample Characteristics

Table 1 presents the dependent and explanatory variables used in the study and their description. Tables 2 and 3 report the summary statistics for continuous and discrete variables, respectively. An average worker in the sample is a 36-year-old male from a rural area, and wage earning of 2508 INR (about 36 USD) per week. By contrast, informal sector workers are 3 years younger, and earn wages that are one-third of that of formal sector workers. On the contrary, those with informal employment are 7 years younger, and earn wages that are less than one-third of those having formal employment.

While the wide gap in educational attainment between formal and informal workers is clear. More than 60% of formal workers have higher than secondary education (completed 10th grade/high school), whereas more than 60% of informal workers have lower than secondary education. In the econometric analysis, education was entered as a continuous variable.¹⁰

There are considerably fewer females in both the informal sector and informal employment sub-samples. In terms of variations by social group, individuals belonging to scheduled caste (SC) and other backward classes (OBC) are concentrated in the informal sector and informal employment. OBCs are preponderant in the informal sector as well as informal employment: 43% of informal sector workers and 42% of informal employment workers are from the OBC category in comparison with their lower representation in the workforce (38%). The average land ownership of informal workers is almost half of that of formal workers. Less than 6% of workers have received formal vocational training in the total workforce, and roughly 4% have received formal vocational training in the informal sector and informal employment.

Results and Discussion

We estimated the determinants of participation in four mutually exclusive employment categories, viz. ISIE, ISFE, FSIE and FSFE, while controlling for selection bias, and state of residence of the individual. Although, endogeneity of one's own education could be an issue in determining choice of the sector, the absence of an appropriate instrument in the dataset restricted us researchers from addressing the issue. Table 4 presents the results of the probit model for labour force participation. As discussed earlier, the IMR from the probit model was used as an explanatory

¹⁰ Results based on education categories are not provided for brevity but can be provided upon request. The main findings of our paper do not change qualitatively when we treat education as a categorical variable.



Table 4 Probit regression for labour force participation

Variable	Employment
Age	0.377*** (0.002)
Age squared	-0.005*** (0.000)
Education (Reference. Not literate)	
Below primary	0.176*** (0.017)
Primary	0.236*** (0.016)
Middle	0.135*** (0.014)
Secondary	-0.055*** (0.015)
Higher secondary	-0.173*** (0.016)
Diploma/certificate	0.434*** (0.035)
Graduate	0.358*** (0.017)
Postgraduate and above	0.835*** (0.027)
Vocational training	0.336*** (0.022)
Female	-2.551*** (0.010)
Social group (Reference. Others)	
ST	0.279*** (0.018)
SC	0.224*** (0.014)
OBC	0.093*** (0.011)
Religion (Reference. Hindu)	
Islam	0.219*** (0.013)
Christianity	0.005 (0.024)
Sikhism	-0.068** (0.035)
Others	-0.037 (0.033)
Rural	0.070*** (0.010)

Table 4 (continued)

Variable	Employment
Log land	−0.073*** (0.002)
DR	−4.240*** (0.026)
State	Yes
Constant	−2.910*** (0.046)
Observations	210,491

Standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

variable in a multinomial logit regression model to control for selection bias. Table 5 presents the results of the multinomial logit model for participation in the employment categories, with FSFE being assigned as the base category. Upon controlling for sample selection, factors such as education, gender, land holding, social group and vocational training played a significant role in determining participation. The IMR is significant in the case of FSIE and ISIE, but insignificant in the case of ISFE.

Education is one of the important factors determining participation in employment. Workers with lower educational attainment were significantly more likely to be employed in both categories of informal work. This is in conformity with the pattern shown in the descriptive statistics (Table 3): workers with education below secondary level are more likely to have informal sector employment, whereas those with higher education attainment are predominantly in the formal sector. Education could possibly be one of the important factors restricting entry into formal sector work, because even within the formal sector, those with formal schooling are less likely to be in FSIE.

In the absence of information on actual work experience, the age of the worker can be treated as a proxy for potential experience. Worker's age is statistically significant in determining participation in all informal employment categories relative to the base category. Worker's age and informality have a negative relationship. This may be arise due to the possibility of workers moving to formal sector after attaining experience in the informal sector (Lehmann 2015; Sahoo and Neog 2017; Abraham 2017).

The gender-based differences in informal work are stark. We found that female workers were significantly less likely to have ISIE. This is in accordance with the pattern described in the summary statistics (Table 3), where even though the share of females in the workforce was considerably lower than that of males, their share in the formal sector was higher than that in the informal sector. It was noted that females were also less likely to be employed in the FSIE. This should be interpreted in the context of the trend of the low and falling female labour force participation rate (FLFPR) in India (Himanshu 2011; Mazumdar and Neetha 2011; Afridi et al. 2016). In addition, 38% of females reported 'attending domestic duties' as their working status in the NSS 68th round, which is not recognised as economic activity.



Table 5 Multinomial logit regression for choice of employment category

Variable	FSIE	ISFE	ISIE
Age	-0.123*** (0.011)	-0.117*** (0.030)	-0.112*** (0.009)
Age squared	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
Education (Reference: Not literate)			
Below primary	-0.487*** (0.083)	0.056 (0.257)	-0.383*** (0.073)
Primary	-0.729*** (0.075)	0.352 (0.219)	-0.547*** (0.066)
Middle	-1.321*** (0.066)	-0.469** (0.213)	-1.188*** (0.057)
Secondary	-2.057*** (0.066)	-0.811*** (0.209)	-1.970*** (0.056)
Higher secondary	-2.587*** (0.069)	-0.616*** (0.204)	-2.653*** (0.057)
Diploma/certificate	-3.113*** (0.096)	-1.266*** (0.280)	-3.562*** (0.075)
Graduate	-3.011*** (0.067)	-0.961*** (0.200)	-3.508*** (0.056)
Postgraduate and above	-3.232*** (0.077)	-1.005*** (0.220)	-4.318*** (0.065)
Vocational training	-0.134** (0.054)	-0.156 (0.151)	-0.372*** (0.041)
Female	0.371*** (0.053)	0.056 (0.159)	-0.194*** (0.044)
Social group (Reference. Others)			
ST	-0.338*** (0.057)	0.269* (0.155)	-0.721*** (0.045)
SC	-0.167*** (0.045)	0.127 (0.132)	-0.520*** (0.035)
OBC	-0.084** (0.037)	0.042 (0.107)	-0.009 (0.028)
Religion (Reference. Hindu)			
Islam	-0.065 (0.050)	0.345** (0.135)	0.197*** (0.038)
Christianity	0.104 (0.075)	0.139 (0.199)	0.013 (0.057)
Sikhism	-0.074 (0.115)	-0.078 (0.341)	-0.017 (0.090)
Others	0.231** (0.108)	0.146 (0.282)	0.508*** (0.078)
Rural	0.219*** (0.032)	0.149 (0.092)	-0.031 (0.024)

Table 5 (continued)

Variable	FSIE	ISFE	ISIE
Log land	−0.085*** (0.007)	−0.070*** (0.021)	−0.084*** (0.006)
State	Yes	Yes	Yes
IMR	−0.292*** (0.048)	−0.201 (0.146)	−0.204*** (0.040)
Constant	5.451*** (0.225)	0.286 (0.642)	6.491*** (0.185)
Observations	81,360	81,360	81,360

Standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Coefficients reported are estimates of multinomial logit model. Relative Risk Ratios (RRR) can be obtained by exponentiating the multinomial logit coefficients

Domestic duties, such as cooking for the household and childcare, are categorised as economic activities when such activities are performed in an employer's household. Similarly, unpaid family members in the households are primarily females.

Females may, however, prefer to work in the formal sector, because the formal sector usually offers provisions such as childcare and maternity leave, unlike in the informal sector. Females' being less likely to work in the informal sector is further indicative of the presence of non-economic disincentives. The participation of women in FSIE can be attributed to non-wage factors, such as cultural norms, presence of children, proximity to home and flexible work hours (Hunt and Samman 2016).

Land ownership plays an important role in determining participation, particularly in the rural context as land can be considered as a proxy for assets and additional sources of income, such as rent. Those owning land can partly work in their own fields, and those with high endowment of land can invest in skills and education. It is by virtue of such human capital investment that they are more likely to find a job in the formal sector. The results of this study suggest that workers with higher land ownership have significantly less chance of being engaged in informal work.

Controlling for variations across social groups, there were significant inter-group differences. STs and SCs were less likely to work in the informal sector compared with those from the unreserved group (general category), but OBCs were more likely to work in the informal sector relative to the unreserved group. It was evident from descriptive statistics that OBCs mostly worked in the informal sector, as around 40% (Table 3). Among social groups having reservations in public sector employment and educational institutes, SCs and STs are preponderant. Therefore, the lower likelihood of SCs and STs in the informal sector may be attributed to these reservations.

Vocational training was associated with significantly lower participation in informal work. Vocational training is an important policy variable, from the perspective of increasing wages and formalization of the workforce. This is reflected in findings of recent studies in literature that have reported the role of skills in the



transformation of informal sector employment to formal sector employment (Mehrotra et al. 2015). Further, growing mechanisation and technical changes in economic activity necessitate the need for a skilled workforce.

There is also evidence of a clear rural–urban divide in participation. Workers in rural areas were less likely to work in informal jobs than their urban counterparts. While large scale public employment programs such as the Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS) offer avenues for informal sector work for casual labourers; rural, non-farm-sector work, particularly in the construction sector, has also increased (Jatav and Sen 2013).¹¹ The lower likelihood of rural workers being in informal work can be attributed to the primary dependence on agricultural activities; that are neither formal nor informal as per definition. Nevertheless, the growing rural non-farm sector is an important source of livelihood for millions, and has a bearing on overall informality of the workforce (Binswanger-Mkhize 2013).

Conclusions

Given the large share of informality in most developing economies such as India, it is important to study the determinants of informality. In this study, we estimated the determinants of informal labour market participation. Using unit-level data from the 68th round of the NSSO employment–unemployment survey (2011–2012) in India, which comprises approximately half a million individuals, we showed how the determinants of informality depended on the definitions of informality used.

Controlling for selection bias in labour force participation, and state of residence of individuals, we found that, irrespective of the definition of informality used, informal jobholders had low educational qualifications. The chances of informality were reduced with education. However, in light of the evidence that educated youth are less likely to be employed — informal or otherwise — along with limited opportunities in the formal sector (Bairagya 2018), formalising the role of education may be seriously restricted. Furthermore, those with formal vocational training were less likely to engage in informal employment. This finding is relevant in the context of growing concerns about the challenges faced by initiatives under the ‘Skill India’ mission, following the implementation of the National Policy for Skill Development and Entrepreneurship 2015 (MSDE 2016). Since the outreach of formal vocational training remains meagre — around 3% of the workforce (Labour Bureau 2017), it is important to address major challenges in vocational education and training (Mehrotra et al. 2015).

We found significant gender variations, with females being less likely to participate in informal work. Given the recent trend of the low and falling female labour force participation rate in India (Mazumdar and Neetha 2011; Mehrotra and Parida

¹¹ The MGNREGS aims to enhance the livelihood security of people in rural areas by guaranteeing 100 days of wage-employment in a financial year to a rural household whose adult members volunteer to do unskilled manual work in the vicinity.



2017), the lower participation of women in informal work raises questions about the provision of critical services such as maternity and childcare support in informal jobs. Furthermore, given the occupational vulnerability of women in the informal sector in particular, measures for protection of women in the workplace are also critical (ILO 2018). Since the decline in the female labour force is largely due to young women opting for education, provisions for vocational training and skills training programmes should be designed keeping in mind the aforementioned vulnerabilities of women workers.

In light of this study's findings and emerging trends of unemployment and informality in India, there is a need to restructure skill development programmes to account for the heterogeneity of informal workers (Radchenko 2014). Moreover, given the potential entry barriers for females who are mothers of young children, provisions for childcare to support women in the informal sector should be made. This is even more pressing in the context of the recent trend of declining female labour force participation rate in general, and rural labour force participation rate in particular.

The few limitations of the study include the fact that we did not address the potential endogeneity of worker's own education. Even though parental education significantly influences the labour market participation of children, due to a lack of data on personal traits and motivators, we were unable to shed light on the nature versus nurture (Behrman and Taubman 1989) aspect of informal employment and informal sector employment. This could be addressed in future studies using richer microdata. Furthermore, the analysis was confined to employment based on UPAS, thereby ignoring subsidiary activities. As India's economy is growing, there has also been a growth of subsidiary activities. It can be hypothesised that the availability of subsidiary activities may influence the decision to participate in informal work, as well as the reservation wages in the informal sector. In future research, subsidiary activities and employment based on different recall periods may also be examined.

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